# EmWoT as a generalized theory of Points of View

## Scope and motivation

> «There's little math can do to address the inadequate political response to the climate crisis, but applied category theory is further along in other areas of public concern.»
> Natalie Wolchover, columnist for Quanta, in her article 'Can the most abstract math make the world a better place?'

In this article Wolchover describes the laudable efforts of a community of mathematicians offering to leverage their expertise in deploying solutions to the environmental debacle. This quote sits exactly in the middle of the article, like a black hole of impossibility around which revolves what this math is, what it can do and what it has already achieved: to model the natural world while respecting its complexity. But this endeavour seems irrational. If Nature has worked perfectly fine for billions of years and humans have suddenly turned against Her, why should science focus on the healthy side of this two-part system instead of trying to figure out the aggression precisely by addressing "the inadequate political response to the climate crisis". If the most abstract math can make sense of Nature, why can't it help in understanding the shortcomings of our collective organization?

Here's the response of a member of this community, John Baez, clarifying their intentions:

> «"I don't think we have the math to understand such [living] systems yet," said Baez, who thinks modeling these systems will involve new categories with previously unstudied logical structures. "That's the kind of mathematics I would like to develop, because I have this hope that it will help us be kinder to the world if we understand the world a bit better and not think of the natural world as raw materials for our machines to take advantage of to do what we want to do. That attitude that we have right now is running into a wall. That attitude winds up destroying the whole planet."»

A collective "we" typical of the polite false naiveté of a hard science community content with an idealized view of a political system which still has human intentions parameterized by understanding, and in line with the desires of the demos—as opposed to being made out of pure strategies to navigate and circumvent the laws of the state and the market, solely governed by the optimization of the exploitation of both Nature and humans. An attitude which allows them to avoid the long due hard look in the mirror, as the main purveyors of the technologies which enable this exploitation.

A "we" this text will suggest can be rearranged and modelled on an understanding of evolution devoid of the hierarchical and inherently violent assumptions stealthily injected in the mainstream view by Herbert Spencer as a justification for what is now known as neoliberalism, or whatever is the even more degenerate broligarchy it has turned into (this follows a path circling back to the enlightenment and ultimately an age old tradition historically known as Scala Naturae). As an alternative to the top-down and tree-like rigid hierarchy of the "survival of the fittest", the text will propose an emergent network architecture of the "befitting of the most connected" (as established in the ecosystem by the success of the mammals, whose distinctive trait is *attachment*—which has a beautiful mathematical counterpart in something called the *Yoneda lemma*). This should produce a "we, the philosophers" capable of influencing the political and economical realm, and which would be in a very real sense the rigorous extension of the "I, the scientific" of

the Galilean scientific method, for now doomed by isolation to passively chronicle the tragedy, one failed COP and alarming GIEC report at a time.

Interestingly, the main architect of the wonderful mathematical abstractions discussed in the article and used in our model is the most striking example of a prominent mathematician entering the political realm: Alexander Grothendieck. Arguably the most important figure of 20th century mathematics, he quit everything at the apex of his career in 1970 to start an ecological movement and magazine, 'Survivre', and in 1972 gave a conference at CERN: 'Will we continue scientific research?'. His efforts were met with contempt from his former colleagues and marked by lack of mutual understanding with his new clique. The rest of his life will be a social disaster and a profusion of writings spanning from mathematical brilliance to esoteric explorations into dreams, arithmetic structure or evil.

In the CERN conference he lays out some foundations for repairing the social and political by starting with the most local of relations, with the everyday conversations, and crucially, by eliminating the hierarchical structures looming over every aspect of everyday life. His first example is between teachers and students. However, he didn't have a concrete plan for it. Converging with the idea of a collective scientific method, this is precisely what the EmWoT implementation presented in this text sets to achieve. And what this author started with, completely independently of Grothendieck and his mathematics. Actually, one of my main motivation at the time was to demonstrate that math can't be used to model society and is extremely dangerous in the hands of economists, because the difference in complexity excludes the "philosophical air" humans can't survive without from the models. (I still believe this is true, only I have revised my judgement on how much complexity mathematics can embrace, when it breathes the same philosophical air we do.)

In retrospect, and suppressing the details of how I got there (I've exposed it in other documents you can find at the end of this one), it is evident that mathematics are way too vast to tell you what the optimal social structure is: it is very eager to tell you an infinite number of beautiful stories which all seem to perfectly match your wants, but all equivalently so. This is precisely the problem string theory has: too many candidate solutions and not a single criterion to distinguish between them (something John Baez and his friends know very well, and probably the real reason why they avoid weighing in on the politics). On the other hand, what I have found in my research is that if you engineer a complex enough optimal algorithm for social interaction, one whose very adequateness is the same as what is baked in as invariants of our context by Nature, then the sufficiently holistic mathematics that model this structure will "light up", and illuminate and relate immense regions of knowledge, paving the way for something best labeled as "omnidisciplinarity". The very structure of "how knowledge appears to us" becomes available, as a theory of point of view coupled with a generalized theory of evolution. What you get is not a demonstration in the reductionist sense, but mathematical elegance married with philosophical depth, and a workable plan to implement it in the real world, at scale, and test its results.

## Historical imprints of the generalized notion of health

The main feature of, EmWoT, the theory of point of view, is its notion of health. If we restrict what a point of view can be to organisms with brains, then we can focus on the set of all relations between all the cells, and notably the neurons, and call this a "conversation" (this conversation can be properly modelled using computer science, namely the actor model of concurrency theory). It is this conversation that can be more or less healthy. Not in the normative sense of whether it "agrees on a particular thing", in line with an absolute objective point of view, as most institutions like schools or psychiatric hospitals imagine health to be, but in a loose sense of whether the conversation "resonates" and finds an "equilibrium": it has many moving parts

and is defined not through them but through how they interact, and there are infinitely many ways in which the network of interaction can do its job adequately (deciphering signals, world mapping, information processing…). Like with all complex systems, perfection is not unattainable but a myth, one must navigate trade-offs.

Now, switching to humanity as the point of view of interest, we can locate in its history when it lost its equilibrium, by looking at two types of buildings, the temple and the palace, appearing in close succession in the archeological record when the grip on populations of the newly formed central authorities reached an unprecedented level, "at the dawn of civilization"—I will not engage here in the analysis of the trade-offs "civilization" offers, and refer the interested reader to 'The Role of Women in Work and Society in the Ancient Near East', starting with the the chapter by Bertrand Lafont, and simply ask: if it starts with the generalization of slavery and ends with the destruction of the ecosystem, is it a healthy mode of being?

We can recognize these two buildings as enacting the partitioning and hacking of the equilibrium/resonance of the communication system of human culture, into a specialized "why", the temple, and a specialized "how", the palace. Specialization here means both "expertise" and "special interests", with the former serving as justification for the degeneracies emanating from the latter. Together they manage to do what a virus does to a cell: to repurpose the potential of the system for its own gain, with little regard for the integrity of the cell or its context. Or what is transpiring in human history as the unconstrained exploitation of Nature and the human workforce to complete exhaustion. (Precisely what I claimed the applied category theorists above are ignoring in their diagnosis, leading to an ineffective solution—and it's precisely this community who hold the key to the appropriate solution, because one should read "applied category theory" as the reunification of the "how" and the "why" into a holistic and actionable vision).

Because one's point of view is what induces one's behavior, the "broken conversation" produces a "broken implementation", or economy. What we have now is a two-cycle of: { invest → produce → sell | buy → consume } in which the two sides are separated, the production must be maximal (including through absurd strategies such as planned obsolescence), the supposed benefits at the end of the cycle are a dead end for the flow, only there to validate the original impulse, mostly generating garbage and hoarding, and nothing in there is connected to the ecosystem, except, once again, exploitation. This schema is work first, autonomy second: the individual agent can gain autonomy only through work, not of their choice but from a pool of tasks decided at the investment step, without guarantee that everyone will have access to autonomy. There is enough in the overproduction to satisfy a lot of the basic needs for large portions of the population, but at the cost of freedom and the deterioration of context, notably the social fabric and ecosystem.

A healthy conversation would eventually completely *reverse* this, and produce: { inventories & demands by reservations → communally organized production & distribution → use & release → inventories… }. There is but one loop, which is solidarity first and work second: the loop starts with the assessment of needs, proceeds to use voluntary work and a sensible management of the resources to cover those needs, immediately distributes according to set patterns and after usage the goods return to the inventory. Every step requires a communication system all can trust. Both social science and practice in the field came close to formalizing what they should be (such as citizens' assemblies or various systems resembling corvé), but through trials and errors, and as partial implementations it's not obvious why we should adopt them. Without them and under the pressure of the temples and palaces, the current economy was *forced* to adopt the invest model.

I realize to most readers this dichotomy will read as "the unescapable reality of human behavior" versus an "utopian dream", but if you look at the new vision academic research in human history is converging to when

it sheds the heavy assumptions inherited from centuries of justifications for "civilization", you will understand that it is actually closer to "mistaking the virus for the cell" versus "how humanity functioned for 99% of its history, showed signs of healthy civilization-like growth which were then submerged by neighbors with cancerous growth patterns". In the next section this dyad will resurface as two types of flows in the theory, healthy and virus-like, reflecting the autonomy-first or work-first dynamic. It simply says that a human which is granted autonomy and dignity by their community, without expectations of a return, is then free to chose to engage in complex work, or not. The system works because it doesn't need everyone to chose to engage in complex work, but there will always be enough to take this path for the solidarity system which grants dignity universally to persist, allowing the next generation of complex worker to emerge. Simply because complex work is interesting.

What EmWoT and omnidisciplinarity do is recognize the conversation and human history as an instance of a recurring pattern in the history of the universe, which describes the *attachment* relations between complex systems, and the effects of distance and scale on how they perceive each other. To us this attachment relation is the observation that our distinct world emerged with the biological ecosystem as our "parental figure". Our "grandparent" is chemistry on Earth, itself a descendent of subatomic physics and the big bang. Each step in the progression involves a change in scale and a kind of "truncation of parameters", both quantitatively and qualitatively, such that at the end of the chain, elements can be defined by a few precise numbers and binary parameter (like the spin of atoms). This should be understood as an "optical illusion", an effect of the limitations of our human Earth-bound point of view, not a limitation of what those things are in themselves. We must imagine that if they were to "look at the whole of us", they might perceive a tiny speck of material which is either in meaningless state A or meaningless state B. Besides those perspectival distortions, what EmWoT introduces is a recurring pattern, an echo that runs through the network of complex systems: each new system will "retell the story told by their predecessors", in a new voice and a new language, but preserving some crucial invariants, including the framework which enables the network to create new nodes and expand. In the following section we will unfold this framework in a language spoken by all the nodes, mathematics, with a translation in the ones we should be most familiar with, philosophy.

We can enclose this framework in a very simple diagram:

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> Figure 1:
> CoE stands for "Cone of Entry", to mean the vast history inside causal structures that converges into the formation of distinct Points of View and their internal conversations—something reductionist science is well equipped to study and has done a wonderful job with, but is now stuck because enable to access the complexity of the dual concept of "causal fields", the "perspectivist fields". Conv/PoV stands for the Points of View and their internal and external conversations. Only accessible through omnidisciplinarity, they are the elements through which a causal structure achieves a kind of "Kantian moment", or "factorizes", and participates meaningfully in the last element: G-evol, for Generalized Evolution, the network described briefly in the paragraph just above (and more in details in another text, for which you can find a reference at the end of this text).

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## The theory of Points of View as an icosahedron

### The structure

To bridge the familiar and trusted science of the CoE realm and the seemingly impenetrable territory of G-evol via the philosophically loaded concept of perspectivism, we'll start with a fixed geometry, a Platonic solid: the icosahedron (which is in a rigorous sense "the most simple of the complex shapes"), and:

- Assign a philosophical concept to each of its 12 vertices, the 12 most fundamental concepts needed to reconstruct the theory represented in Figure 1—we will think of those vertices as modules in a complex information processing program.
- Define certain processes on the icosahedron, such as communication patterns, modes or flows, and give them a practical and philosophical meaning.
- "Step back" into the of mathematics of Grothendieck and [embellish] our structure [with blablabla which will give us access to HoTT]
- [homotopy = empathy, type = world]
- Compose a dictionary that translates the 12 philosophical concepts into 12 fundamental types of homotopy type theory, preserving the connection to the target framework G-evol.
- Use 8 other fundamental types of HoTT, unused by the vertices, to model the conversation as an implementation of the actor model of concurrency theory inside the icosahedron (both as local conversations at each vertex and as a global conversation involving all vertices).
- Give a practical philosophical meaning to those 8 types in order to replicate the communication system on the philosophical side.
- Utilize the dictionary to test the construction structurally and numerically in three layers of complexity: the statics, the uniform dynamics (which respects the A_5 symmetry of the icosahedron), the perturbative dynamics (the processes which break this symmetry), using a proof assistant (Lean or Agda) and dedicated programming languages (Haskell, Python).

If you do one thing on one side, the dual gives you a meaningful counterpoint (ex: HoTT -> φ gives a conversation toolkit which wasn't obvious on this side, φ -> HoTT)

## The 12 concepts

In philosophy
In HoTT
In engineering as generalized permaculture (what is it)

## The conversation

Eventually we can construct a real communication system for humans, an implementation of the collective scientific method, which takes as input any number of humans (preferably in the hundreds or thousands), and outputs a documentation and a specification for a technical language for this documentation. The algorithm is as follows:

- The unit element of the system is a conversation (in Linux everything is a file, in lambda calculus everything is a function, in)
- system creates "seeding" conversations -> define a concept to explore
- system creates an icosahedron of 12 conversation for this concept (all tightly connected in a global conv, with possible connection to other global or local conversations in the universe of concepts)
- actor model pattern: send, receive, update state by consensus, create
- the total documentation is all the states and homotopies

The "we" of Baez versus this "we", if infused with the intelligence seen in citezens' assemblies and the energy seen in 2019 popular protests (the most "sheaf theoretic" social phenomena of the last decade)
How the system transforms individually assigned expertise to documentation assigned expertise (how it doesn't make people any less knowledgeable, and the language clearer by removing the incentive for expert domains)

### The processes

- Help, target, align
    - In philosophy
    - In HoTT
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Note:
For those who can read this kind of language, the details of the mathematical implementation are available in Appendix A at the end of the document.

- Statics - 12 vertices (ecosystem analogy, trade-offs):
    - As philosophical modules (labels & definitions)
    - As types in HoTT (labels & definitions)
    - In conv, edges, faces, paths = message-passing (common to both); HoTT → φ = homotopy → (bits of) empathy
- Uniform dynamics - the symmetry group A_5:
    - Help neighbors / align with antipodes (antipodal axes as "fundamental group": univalence, anchor of meaning, fractal growth engine, spectral dimensional ladder…)
    - Awake mode: 12 vertices as different angles for looking at a map (world navigation)
    - Dream mode: 12 vertices as local conversations participating in a global conversation (cubical HoTT, introspection/map update/training)
    - The 8 types/modules enabling the conversation (actor model, unityped empathy/adapter layer)
- Perturbations - the qualitative carving (finding optimality/bottoms of energy wells)
    - Target distance 2
    - Role of infinity, release/infinitesimals, consensus (universe, truncation and A/R) in switching between the two modes/breathing (Jedrezjewski, modal type theory)
    - Self political flow, its Environment counterpart (as triangles and hemispheres)
    - 5 groupings based on golden triangles
- Testing:
    - Mathematics: 3 levels implementation, overview of the construction, modularity in all perturbations, modularity in test suites
    - Social: the same structure as a real open conversation, implementation overview

## Related structures

- Cosmic flower, Spinozism (univocity of Being), actor model, attachment, contextuality, ecosystem (EES) → listening to Nature
- Fractal and conformal systems, bulk/boundary correspondences, quantum gravity → Deleuze's outburst: physics needs metaphysics (omnidisciplinarity)
- Autopoiesis, (M/R), Ehersman & V, neuroscience and theories of mind, cybernetics

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## A new look at the anthropic principle

- Anthropic principle (all causal fields are logically warped by the presence of observers) as dual of the relation between autonomous agent and its horizons (all perpectivist fields are configured by their limits)
- If physics is finely-tuned for a "staircase of complexity" with us at the top, it's an optical illusion of G-evol, from a local stance (concerns a few constants)
- If math (HoTT) is structured as a cognitive engine, it's the same optical illusion, from a global stance (concerns the entire structure) — obvious from an omnidisciplinary perspective
- If both are about information processing, it's because we are information processing agents, and the world/our horizon will appear to us as information processing

## Further reading

- Grothendieck's generalized theory of evolution (the G-evol in CoE → Conv → PoV → G-evol)
- Liberation Networks

// mystery ~ art ~ immanence (God is not beyond but in all things)